

Efficient Genomic Classification with Retrieval-Augmented Approach

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Abstract

Large pre-trained DNA language models such as DNABERT-2, Nucleotide Transformer and HyenaDNA have shown strong performance across diverse genomic benchmarks. However, most applications rely on computationally expensive fine-tuning, which performs best when the training and test data distributions are similar. In this work, we explore whether task-specific fine-tuning is always necessary. We demonstrate that simple embedding-based pipelines—extracting fixed representations from a DNA language model and feeding them into lightweight classifiers—can achieve competitive performance. Here, we focus on HyenaDNA for embedding extraction.

In settings with independent data distributions, embedding-based methods using HyenaDNA perform very well. Our results indicate that embedding extraction provides not only a strong baseline but also a more generalizable and efficient alternative to fine-tuning. Qualitatively, a retrieval-augmented pipeline combining fixed HyenaDNA embeddings with lightweight sequence features achieves good results across three genomic tasks. Additionally, we ran machine learning classifiers on one of the datasets, including Naive Bayes, Logistic Regression and XGBoost (XGB).

1 Introduction

Genomic sequence analysis is a key focus in bioinformatics, aimed at identifying functional DNA elements like promoters, enhancers, and splice sites. Understanding these elements is crucial for insights into gene regulation, disease prediction, and personalized medicine. Traditional computational methods, which rely on handcrafted features or predefined models, often struggle to capture the complex patterns within DNA sequences. Transformer-based models, adapted from natural language processing, have achieved impressive performance on genomic tasks, but their high computational demands and long training times limit practical use. Solving this problem is important for advancing genomic research, enabling faster and more cost-effective analyses in computational biology.

The first benchmarking models in genomic foundation models had compared common models of architecture as DNABERT, DNABERT-2, Nucleotide Transformer (NT) and the more recent HyenaDNA model. The models were evaluated within the context of 57 datasets and specialization of models and tasks in their study of Benchmarking DNA Foundation Models of Genomic Sequence Classification: DNABERT-2 was shown to be more effective with human genome classification, NT-v2 with epigenetic predictions, and HyenaDNA with better runtime scaling and ability to work with long sequences. Among the principal methodological findings was the fact that the average pooling of embeddings was superior to the sentence-level tokens by up to 9.7 more AUC. The work, however, only went as far as classification work and zero-shot evaluation, and regression problems and parameter efficient fine-tuning, as well as biological interpretability, had not been examined thoroughly [1].

Germination of the weaknesses of such strategies as tokenization were also explored in Exploring the Embedding Methods in Genomic Language Models where DNABERT-1, DNABERT-2, NT and HyenaDNA were contrasted in benchmarks of genomic understanding, and a custom phage-versus-bacteria task. It was discovered that tokenization through k-mer is more adaptable to shorter sequences, whereas BPE (DNABERT-2) and nucleotide-based embeddings (HyenaDNA) can be generalized to longer sequences. DNABERT-2 achieved

the highest phage identification ($MCC = 0.96$) but HyenaDNA had less variation in replications hence showing that it is a stable training [2]. However, some issues persisted concerning reproducibility and the difference in vocabulary between inter-species (38.4% overlap of bacterial and mammalian BPE tokens) showed how difficult it was to come up with the universal-strength genomic frameworks. These findings suggest that, in order to transfer models to other tasks and species, design, be it k-mer, BPE, or nucleotide-level, is necessary.

The greatest innovation was the introduction of HyenaDNA, a long-range genomic LM that tries to solve the quadratic scaling of Transformers by replacing attention with implicit convolutions. HyenaDNA was a one-base-resolution sequencer with a scale of modeling a million-base-long sequence in model at any given moment, which was 500 times better in context length than its predecessors. HyenaDNA (has 1,500 fewer parameters, and 3,200- less training data than the Nucleotide Transformer) yielded state-of-the-art performance on 12 of 18 NT benchmarks and 7 of 8 GenomicBenchmarks. It was also the first to learn in-context in genomics, presenting soft prompt tokens, which display adaptation to new tasks without gradual weight changes [3]. Nevertheless, the model was trained exclusively on the human reference genome which can limit its cross-species generalization and its in-context experiments were limited to simpler tasks of adaptation. The reproducibility problems were observed in the external assessments, which implies that the methodological improvements are to be further developed to guarantee the stability of the methodology across the different run and datasets.

The retrieval-augmented approaches based on NLP provided an alternative perspective on the outputs of the underlying grounding models. The first model introduced was the Retrieval-augmented generation (RAG) [4], and it consisted of a heavy retriever and a BART generator and reduced hallucinations by 42 points on knowledge-intensive NLP tasks. As much as it has been established as working well in the textual areas, RAG cannot be employed in genome-scale applications, where text can be millions of bases, and 1k-token windows and quadratic attention are used.

GeneRAG augmented retrieval to the genetics field by querying NCBI genome records and prompting GPT-4, improving query-based gene-question accuracy and cell-type annotation by 39 and 43 points [5], respectively, relative to GPT-4. However, GeneRAG continued to model DNA as the short text fragments and long-range genomic dependencies were not modeled. It was also biased to annotation and in contrast with HyenaDNA that oversees learning internally, was founded on curated databases.

Finally, to make long-range modeling strictly evaluated, DNALongBench [6] offered a benchmark set of activities to optimize enhancer-target interactions, eQTL predicting, 3D chromatin contacts, and transcriptional initiation signals activities with maximum context requirements of 1M bp. HyenaDNA was more successful at these tasks, producing an AUROC of 0.828 vs 0.797 at enhancer prediction and is 160x faster on million-scale sizes of attention compared to Transformer-based attention. However, HyenaDNA remained inferior to handcrafted expert models with the use of the biological priors that doubled its results in chromatin contact prediction, and quintupled its correlation scores in transcription signal prediction. Also, Transformer-based genomic LMs were not applied due to the limitations in computing power, open-ended questions of HyenaDNA comparative advantage over good attention-based variants.

Although transformer embeddings can group functionally related sequences, few studies compare retrieval-based classification across models and tasks. We address this by evaluating HyenaDNA within a unified retrieval framework, enhanced with biologically relevant features, offering an efficient and accurate alternative to full model fine-tuning.

1.1 Our Contribution

The key contributions and steps carried out in this project are summarized as follows:

- Used HyenaDNA pretrained embeddings as fixed sequence representations.
- Augmented embeddings with handcrafted genomic features (GC content, z-curve, AT/GC ratio, cumulative skew, pseudoKNC).

- Indexed embeddings with FAISS (L2 distance).
- Applied k-nearest-neighbor retrieval with majority/weighted voting.
- Evaluated on Human Enhancers Cohn, Human Enhancers Ensembl, and Demo Coding vs Intergenic Sequences datasets.
- Ran machine learning models: XGBoost, Logistic Regression, Naive Bayes on Demo Coding vs Intergenic Sequences datasets.

2 Material and Methods

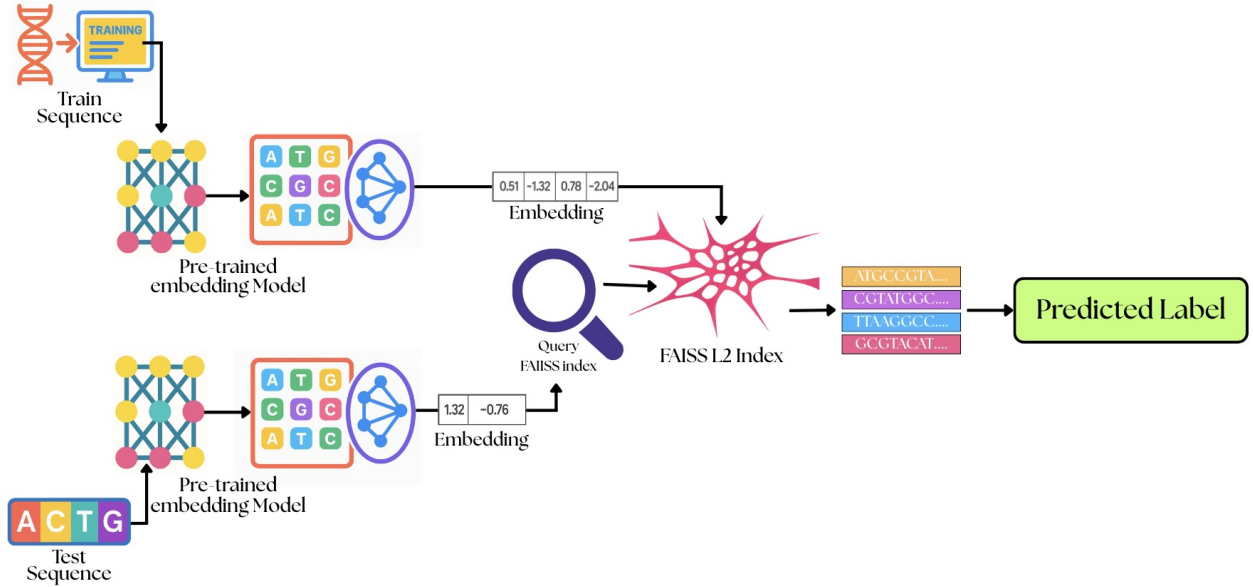


Figure 1: Proposed framework

We introduce a retrieval-augmented genomic classification framework that avoids fine-tuning large models. DNA sequences are embedded using the pretrained HyenaDNA model without parameter updates. During inference, relevant training embeddings are retrieved via nearest-neighbor search and combined with the test sequence. Classification is performed by comparing similarity scores, removing the need for additional layers or retraining. This lightweight approach supports zero-shot or few-shot learning, making it effective for imbalanced datasets. The base framework comprises three components: (1) hybrid feature extraction, (2) FAISS-based retrieval, and (3) label prediction via weighted voting, achieving competitive performance without training complex models.

2.1 Dataset Description

We evaluate our models on genomic benchmark datasets, specifically Human Enhancers Cohn, Human Enhancers Ensembl, and Demo Coding vs Intergenic Sequences, which cover a variety of biological classification tasks. These datasets include enhancer detection, promoter classification, and other regulatory sequence identification tasks across multiple organisms such as humans, mice, worms, and Drosophila. Each dataset provides DNA sequences as input features and categorical labels indicating the functional class of each sequence. We selected these datasets because they are diverse, well-curated, and widely used in genomic research, allowing us to assess the generalizability and robustness of our retrieval-augmented classification framework. All datasets are obtained from the Genomic Benchmark paper [7].

Dataset	Train Sample	Test Sample
Demo Coding vs Intergenic Seqs	75,000	25,000
Human Enhancers Cohn	20,843	6,948
Human Enhancers Ensembl	123,872	30,972

Table 1: Summary of Genomic Datasets

2.2 Tools / Models / Algorithms Used

HyenaDNA Embeddings

HyenaDNA embeddings are the numerical representations of DNA sequences produced by the HyenaDNA model. Instead of treating DNA only as raw symbolic strings (A, T, G, C), HyenaDNA converts them into high-dimensional vectors that capture both local motifs (short-range dependencies like codons, transcription factor binding sites) and global context (long-range regulatory interactions across thousands of bases). Its embedding process leverages Hyena’s hierarchical filtering mechanism, which efficiently integrates convolutional operations for nearby context with implicit operators that model distant relationships. As a result, HyenaDNA embeddings are rich, scalable representations of genomic sequences, making them highly effective as input features for downstream tasks such as enhancer classification, gene regulation prediction, or variant effect analysis. These embeddings not only compress DNA into a machine-readable form but also encode biological structure and dependencies in a way that standard one-hot or k-mer embeddings cannot.

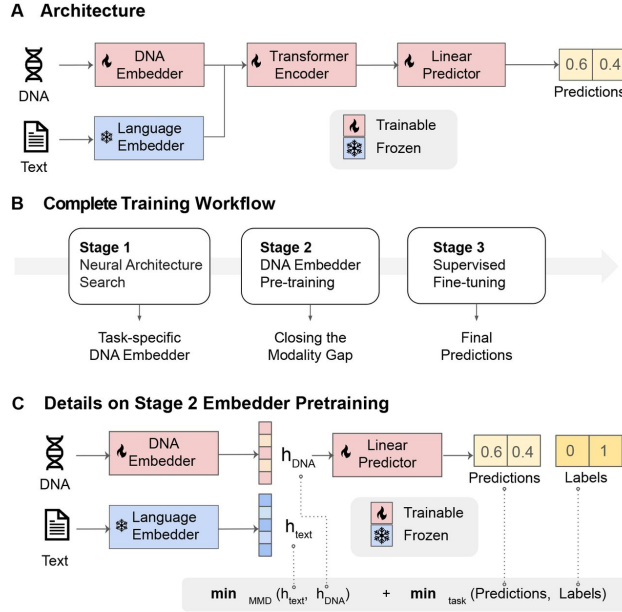


Figure 2: HyenaDNA Embedding

Handcrafted Genomic Features

To enrich sequence representations, several biologically meaningful handcrafted features were extracted:

- **GC Content:** This feature quantifies the proportion of guanine (G) and cytosine (C) nucleotides in a DNA sequence, reflecting its stability and structural characteristics. Regions with high GC content are often associated with coding sequences, promoters, and other functionally significant genomic elements.
- **Z-Curve:** The z-curve provides a unique three-dimensional representation of a DNA sequence, capturing cumulative distributions of A, T, G, and C nucleotides. It effectively encodes global compositional asymmetries, enabling the identification of functional regions such as replication origins and coding sequences.

- **AT/GC Ratio:** This feature measures the relative abundance of adenine-thymine (A-T) pairs to guanine-cytosine (G-C) pairs. Variations in this ratio can indicate sequence-specific structural properties, influence DNA melting temperature, and correlate with regulatory elements and nucleosome positioning.
- **Cumulative Skew:** Cumulative nucleotide skews, such as GC skew or AT skew, capture the imbalance between complementary bases along a sequence. These skews highlight strand-specific asymmetries and are informative for locating replication origins, transcriptional directionality, and other sequence-dependent functional elements.
- **Pseudo K-tuple Nucleotide Composition (pseudoKNC):** PseudoKNC extends traditional k-mer analysis by incorporating both local sequence composition and global sequence-order correlations. This feature captures higher-order dependencies within the sequence, enabling more nuanced representation of structural and functional patterns beyond simple nucleotide frequencies.

FAISS (Facebook AI Similarity Search)

FAISS is a high-performance library developed by Facebook AI for efficient similarity search and clustering of dense vectors. It is designed to handle large-scale datasets, enabling rapid nearest-neighbor search in high-dimensional spaces. By indexing embeddings using structures such as flat L2 indices, inverted files, or quantized representations, FAISS significantly accelerates retrieval while maintaining high accuracy. In genomic sequence analysis, FAISS can be used to index sequence embeddings and perform fast k-nearest-neighbor retrieval, allowing the identification of sequences with similar functional properties without exhaustive pairwise comparisons. Its flexibility and scalability make it particularly useful when combining learned embeddings with handcrafted features for large-scale genomic datasets.

Machine Learning Models

XGBoost:

XGBoost (Extreme Gradient Boosting) is an ensemble learning method that builds multiple decision trees sequentially, where each new tree focuses on correcting the errors of the previous ones. In the context of DNA sequence analysis, XGBoost can use engineered features such as k-mer frequencies, GC content, or embeddings (from models like HyenaDNA) to classify sequences (e.g., enhancers vs. non-enhancers). Its strength lies in handling high-dimensional and sparse genomic data effectively while avoiding overfitting through regularization. Because DNA features often have non-linear relationships, XGBoost's ability to model complex feature interactions makes it a strong baseline and competitive performer in genomics prediction tasks.

Logistic Regression:

Logistic Regression is a simple yet powerful linear classifier that models the probability of a sequence belonging to a particular class using a sigmoid function. For DNA analysis, logistic regression can be applied to predict binary outcomes such as whether a given sequence is an enhancer, a promoter, or disease-associated. For instance, features like motif scores, CpG density, or embeddings can be input into the model, which then assigns weights to each feature to determine how strongly it contributes to the outcome. While Logistic Regression cannot naturally capture non-linear patterns in DNA data, it remains widely used for its interpretability, making it easier to understand which genomic features are most predictive.

Naive Bayes:

Naive Bayes is a probabilistic classifier that applies Bayes' theorem under the assumption that features are conditionally independent. In DNA sequence classification, this means treating sequence features such as k-mer counts, motif presence, or positional nucleotide frequencies as independent evidence contributing to the final classification. Despite this simplifying assumption (which is rarely true in biological data), Naive

Bayes often performs surprisingly well, especially on smaller datasets where more complex models may overfit. It is also computationally efficient, making it suitable for large-scale screening of DNA sequences before applying more sophisticated models.

Parameters and Hyperparameters:

DNA sequences are converted into k-mer representations (k=6) and tokenized with a maximum input length of 512, with labels mapped and datasets formatted as PyTorch tensors. Input sequences are processed in batches via a DataLoader for efficient GPU usage, and token-level outputs from the frozen pre-trained HyenaDNA model are mean-pooled to obtain fixed-length embeddings. Brute-force distance computation is replaced by a FAISS IndexFlatL2 index for faster k-NN retrieval, with inference time tracked to measure efficiency gains. All hybrid models use a k-NN classifier on the combined vector. The current implementation performs zero-shot feature extraction with frozen model parameters, eliminating the need for gradient-based training.

For machine learning models, K-mer features were used with a K-mer size of 5, a maximum of 500 features, and a token pattern of r['_+'] for CountVectorizer. XGBoost (XGBClassifier) was tuned with grid search parameters including max_depth values of 3, 5, and 7, learning_rate values of 0.01, 0.1, and 0.2, n_estimators of 100 and 200, and subsample values of 0.8 and 1.0, with fixed parameters random_state=42 and eval_metric='logloss'. Logistic Regression was tuned with grid search parameters C values of 0.01, 0.1, 1, 10, and 100, penalty options 'l1' and 'l2', and solver='liblinear', with fixed parameters random_state=42 and max_iter=1000. Naive Bayes (GaussianNB) was used with the default parameters from scikit-learn without hyperparameter tuning.

2.3 Performance Evaluation

For HyenaDNA, we have used the original train and test set from the benchmark dataset. For machine learning models, Training Set: 80% of the original training data Validation Set: 20% of the original training data Test Set: The original test set from the benchmark dataset. To assess the performance of our models on genomic sequence classification, we used the following metrics:

Accuracy: Accuracy is one of the most intuitive metrics in machine learning, representing the overall correctness of a model. It is calculated as the proportion of all correctly predicted instances (both true positives and true negatives) to the total number of instances. Accuracy gives a quick sense of performance, but it can be misleading when dealing with imbalanced datasets, where one class dominates. In such cases, a model could achieve high accuracy by simply predicting the majority class, while failing to identify minority class instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: Precision measures the reliability of positive predictions. It is defined as the ratio of true positives to all instances predicted as positive (true positives + false positives). High precision indicates that when the model predicts a positive outcome, it is usually correct. This metric is crucial in applications where false positives carry a high cost, such as medical diagnosis (misdiagnosing a healthy person as sick) or spam detection (marking legitimate emails as spam).

$$Precision = \frac{TP}{TP + FP}$$

Recall: Recall, also known as sensitivity or the true positive rate, measures the model's ability to capture all relevant positive instances. It is calculated as the ratio of true positives to the total number of actual positive instances (true positives + false negatives). High recall is essential in situations where missing a

positive instance is critical, for example, detecting cancer patients or fraudulent transactions. A model with low recall may overlook many true positives, which can have severe consequences.

$$Recall = \frac{TP}{TP + FN}$$

F1-score: F1-score is the harmonic mean of precision and recall, providing a single metric that balances both false positives and false negatives. Unlike simple accuracy, the F1-score is especially useful in imbalanced datasets, as it penalizes extreme disparities between precision and recall. A high F1-score indicates that the model not only identifies positives effectively but also keeps false alarms low, making it a more comprehensive measure of a model’s predictive performance.

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

3 Experimental Analysis

Below are the results showing performance metrics for HyenaDNA with different feature engineering approaches across three genomic datasets (Human Enhancers Ensemble, Demo Coding vs Intergenic Sequences, and Human Enhancers Cohn) followed by performance metrics comparing three machine learning models on the Demo Coding vs Intergenic Sequences dataset.

Human Enhancers Ensemble:

Feature Set	Accuracy	Precision	F1-Score	Recall
Embedding	0.7038	0.7053	0.7033	0.7038
+GC Content	0.6674	0.6849	0.6593	0.6674
+Z-Curve	0.6437	0.6562	0.6364	0.6437
+ATGC Ratio	0.7207	0.7241	0.7196	0.7207
+Cumulative Skew	0.6292	0.6299	0.6287	0.6292

Table 2: Performance comparison across different feature sets (Human Enhancers Ensemble dataset).

Demo Coding vs Intergenic Sequences:

Feature Set	Accuracy	Precision	F1-Score	Recall
Embedding	0.8249	0.8257	0.8248	0.8249
+GC Content	0.7756	0.7904	0.7726	0.7756
+Z-Curve	0.7844	0.7926	0.7829	0.7844
+ATGC Ratio	0.8305	0.8318	0.8303	0.8305
+Cumulative Skew	0.7816	0.7925	0.7796	0.7816
+Pseudo KNC	0.8332	0.8339	0.8331	0.8332

Table 3: Performance comparison across different feature sets (Demo Coding vs Intergenic Sequences).

Human Enhancers Cohn

Feature Set	Accuracy	Precision	F1-Score	Recall
Embedding	0.6684	0.6687	0.6682	0.6684
+GC Content	0.6503	0.6860	0.6326	0.6503
+Z-Curve	0.6826	0.6830	0.6825	0.6826
+ATGC Ratio	0.6782	0.6789	0.6779	0.6782
+Cumulative Skew	0.6753	0.6777	0.6742	0.6753
+Pseudo KNC	0.6681	0.6683	0.6680	0.6681

Table 4: Performance comparison across different feature sets (Enhancer Cohn dataset).

These tables present performance comparisons of machine learning models across three different genomic datasets using various feature engineering approaches. The Human Enhancers Ensemble dataset shows relatively modest performance with accuracy scores ranging from 0.62-0.72, where adding ATGC ratio features provides the best improvement. The Demo Coding vs Intergenic Sequences dataset demonstrates much stronger performance overall, with accuracies between 0.78-0.84, and the ATGC ratio feature addition again yielding the highest scores across all metrics. The Human Enhancers Cohn dataset shows intermediate performance with accuracies from 0.65-0.68, where the Pseudo KNC feature addition performs best. Across all datasets, the baseline embedding features consistently show the lowest performance, while different advanced feature engineering techniques (+ATGC Ratio, +Pseudo KNC) provide optimal improvements depending on the specific dataset characteristics.

Machine Learning Models:

Model	Accuracy	Precision	F1-Score	Recall
XGBoost	0.8846	0.8847	0.8846	0.8846
Logistic Regression	0.8815	0.8815	0.8815	0.8815
Naive Bayes	0.7799	0.7800	0.7799	0.7799

Table 5: Performance comparison across different feature sets (Enhancer Cohn dataset).

This table compares the performance of three machine learning models on the Enhancer Cohn dataset. XGBoost demonstrates the strongest performance with an accuracy of 0.8846 and consistently high scores across all metrics (precision, F1-score, and recall all at 0.8846-0.8847). Logistic Regression performs nearly as well with an accuracy of 0.8815 and identical scores of 0.8815 across all evaluation metrics. Naive Bayes shows notably lower performance with an accuracy of 0.7799 and consistent scores of 0.7799-0.7800 across all metrics. The results suggest that both XGBoost and Logistic Regression are similarly effective for this genomic classification task, significantly outperforming the Naive Bayes baseline.

GitHub Repository: Efficient Genomic Classification with Retrieval-Augmented Approach

4 Conclusion

In this study, we evaluated the effectiveness and efficiency of combining pretrained embeddings with biologically inspired handcrafted features for genomic sequence classification. Our results show that this approach achieves high performance while significantly reducing inference time. Although handcrafted features are static and pretrained embeddings lack adaptability, this design provides substantial computational efficiency without sacrificing accuracy. Moving forward, we aim to develop hybrid approaches that leverage both fixed embeddings and task-specific fine-tuning to enhance adaptability, and to apply this framework to more complex regulatory genomics problems, such as predicting enhancer-promoter interactions and performing multi-label regulatory element classification.

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